American Computer Science League

2022-2023 ● Contest 3: Short Problems ● Senior Division

1. Boolean Algebra

$(\overline{A} + \overline{B})(B + \overline{C}) + \overline{A\,B\,C}$
$= \overline{A}B + \overline{A}\,\overline{C} + \overline{B}B + \overline{B}\,\overline{C} + \overline{A} + \overline{B} + \overline{C}$
$= (\overline{A}B + \overline{A}\,\overline{C} + \overline{A}) + (\overline{B}B + \overline{B}\,\overline{C} + \overline{B}) + \overline{C}$
$= \overline{A} + \overline{B} + \overline{C}$

So $\overline{A} + \overline{B} + \overline{C} = 0\ implies\ \overline{A} = 0,\ \overline{B} = 0,\ \overline{C} = 0.$
Therefore, (1, 1, 1) makes it FALSE. Only 1 works.

D. 1

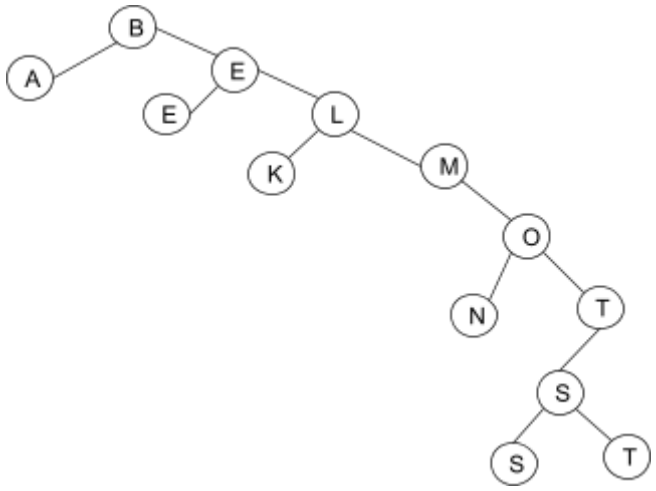
2. Boolean Algebra

$(A \oplus BC)(\overline{A}B \oplus C)$
$= (A\overline{BC} + \overline{A}BC)(\overline{A}B\overline{C} + \overline{\overline{A}BC})$
$= (A(\overline{B} + \overline{C}) + \overline{A}BC)(\overline{A}B\overline{C} + \overline{\overline{A}BC})$
$= (A\overline{B} + A\overline{C} + \overline{A}BC)(\overline{A}B\overline{C} + \overline{\overline{A}BC})$
$= (A\overline{B} + A\overline{C})(\overline{A}B\overline{C} + (A + \overline{B})C)$
$= (A\overline{B} + A\overline{C})(\overline{A}B\overline{C} + AC + \overline{B}C)$
$= A\overline{B}AB\overline{C} + A\overline{B}AC + A\overline{B}\,\overline{B}C + A\overline{C}\,\overline{A}B\overline{C} + A\overline{C}AC + A\overline{C}\,\overline{B}C$
$= 0 + A\overline{B}C + A\overline{B}C + 0 + 0 + 0$
$= A\overline{B}C$

A. $A\overline{B}C$

3. Data Structures

The binary search tree for BELMONTSTAKES is:



The internal path length is: 2*1+2*2+2*3+1*4+2*5+1*6+2*7
= 2+4+6+4+10+6+14 = 46

B. 46

4. Data Structures

The queue is constructed using FIFO as follows:

J, JU, JUL, UL, L, LI, LIU, IU, U, US, S, SC, SCA,
SCAE, CAE, AE, AES, ES, S, SA, A, AR, R

The next item popped is R.

E. R

5. FSAs and Regular Expressions

The following do not satisfy the given regular expression.

[a-m]*[^emtsy]*t.*y

1. crusty fails at s
2. documentary fails at m
3. slapstick fails at first s

B. 3